PROJECT TITLE

SSH BRUTE-FORCE DETECTION USING SPLUNK

Project Domain : **Security Operations Center (SOC)**

Technology **: Splunk Enterprise / SIEM**

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Course : **SOC L1**

Institution : **ADITYA DEGREE & PG COLLEGE**

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INTRODUCTION

A. WHAT IS SSH?

- Secure Shell (SSH) is a network protocol used to securely access and manage a computer or server over a network. It provides an encrypted connection, allowing users and administrators to log in remotely and execute commands. SSH commonly uses authentication methods such as passwords or cryptographic keys to verify a user's identity.

B. What is a brute-force attack?

- A brute-force attack is an attempt to gain access to an account by repeatedly trying different passwords or authentication credentials. In an SSH brute-force attempt, an attacker may make many login attempts against a server, often using different passwords or usernames. Repeated failed login events can be a sign of this activity, although legitimate users may also make several unsuccessful attempts.

C. What is a SOC?

- A Security Operations Center (SOC) is a team or function responsible for monitoring an organization's systems, identifying suspicious activity, investigating security alerts, and responding to potential incidents. SOC analysts review logs and other security data to determine whether activity is normal or potentially malicious. Detecting repeated failed SSH logins is one example of a task that may support this work.

D. What is Splunk?

- Splunk is a platform used to collect, search, and analyze machine-generated data, including system and security logs. Its Search Processing Language (SPL) allows analysts to filter events, count activity, group events by fields such as source IP address or username, and examine patterns over time.

E. Why is this project important?

- This project focuses on using Splunk to examine SSH authentication logs and identify patterns that may indicate brute-force activity. The work includes exploring the available log data, searching for failed login attempts, grouping repeated attempts, and reviewing their

frequency and timing. The project is important because it provides practical experience with log analysis and a basic security-monitoring workflow used in SOC environments.

PROBLEM STATEMENT

SSH servers may receive repeated unsuccessful login attempts when someone tries to guess account credentials. If these attempts are not noticed and investigated, they may indicate a possible attempt to gain unauthorized access. Reviewing authentication logs manually can be time-consuming, especially when the logs contain many events.

This project addresses the problem by using Splunk to search and analyze SSH authentication logs, identify repeated failed login attempts, and highlight patterns that may require further investigation. The analysis is intended to support basic security monitoring; repeated failures alone do not prove that an attack occurred.

OBJECTIVES

The objectives of this project are to:

1. Understand the structure and key fields of SSH authentication logs.
2. Import or access the SSH log dataset in Splunk and confirm that the events are searchable.
3. Use Search Processing Language (SPL) to find failed SSH login attempts.
4. Count and group failed attempts by relevant fields, such as source IP address and username, where those fields are available.
5. Examine the frequency and timing of repeated login failures.
6. Develop basic detection logic to identify activity that may be consistent with an SSH brute-force attempt.
7. Document the queries, results, observations, and limitations of the analysis.

SCOPE OF THE PROJECT

This project focuses on analysing an available SSH authentication log dataset using Splunk. It includes searching for failed login events, examining repeated attempts, grouping events by available log fields, and reviewing activity over time. The project may also examine successful login events following repeated failures if the dataset contains the necessary information.

The project is limited to log-based analysis in a controlled learning environment. The project includes a controlled Hydra simulation against a designated target VM in a private lab, followed by log-based analysis in Splunk. Findings will depend on the contents and quality of the supplied dataset. Any detection threshold used in the project will be treated as a basic investigation rule, not a universal indicator of an attack.

TOOLS AND TECHNOLOGIES

TOOL	PURPOSE
Kali Linux (Target VM)	Used to host the SSH service and generate authentication logs
Kali Linux (Attacker VM)	Used to perform the controlled SSH login simulation
Splunk Enterprise	Used to search and analyse authentication logs
SSH authentication logs(auth.log)	Dataset used for the investigation
SPL	Used to search, filter and analyse events
Microsoft Word	Project documentation

LAB ENVIRONMENT AND DATASET

A. LAB ENVIRONMENT

The project was carried out in a controlled learning environment using Splunk to search and analyse SSH authentication logs. The lab environment consisted of the following components:

- **Operating system:** Kali Linux
- **Log analysis platform:** Splunk
- **Log source:** SSH authentication log file
- **Search language:** Splunk Search Processing Language (SPL)
- **Environment type:** Local lab environment

Splunk was used to search the log events, examine relevant fields, and identify patterns of repeated failed SSH login attempts. The analysis was limited to the log data available in the lab.

B. DATASET

The dataset used in this project was the auth.log file collected from the target Kali Linux machine. This log records authentication-related events, including SSH login attempts and failed password entries.

During the controlled lab simulation, the attacker machine generated SSH login attempts against the target machine. The target's auth.log was then ingested into Splunk for analysis. The log data was used to identify failed SSH login events, extract the source IP address and attempted usernames, and review the frequency and timing of the events.

Dataset source: Target Kali Linux machine
Log file: auth.log
Log type: Linux authentication log
Use in project: Analyse SSH authentication failures in Splunk

C. DATASET LIMITATIONS

The dataset represents only the activity recorded in the provided log file. It may not include all events from a real SSH server, and the available information may be insufficient to determine whether a login attempt was malicious. Therefore, the project identifies patterns that may require investigation rather than confirming that an attack occurred.

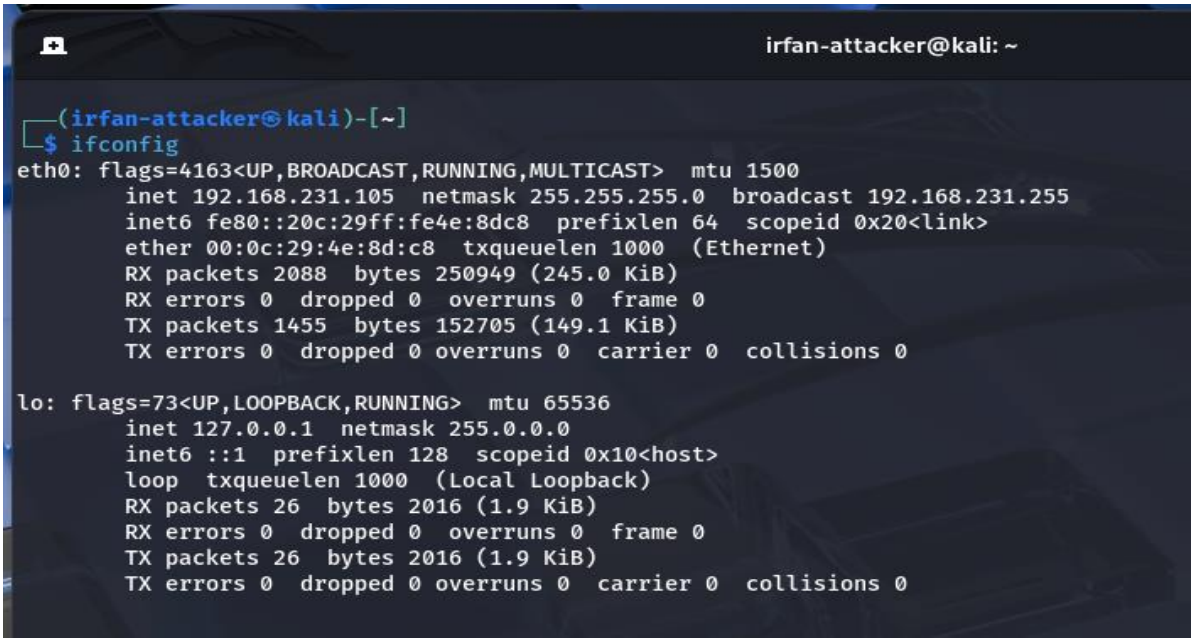
Methodology / Practical Implementation

The project was conducted in a controlled virtual lab to examine SSH authentication activity and investigate repeated failed login attempts using Splunk. The attacker and target were Kali Linux virtual machines. Splunk was used to search and analyse the target's SSH authentication logs.

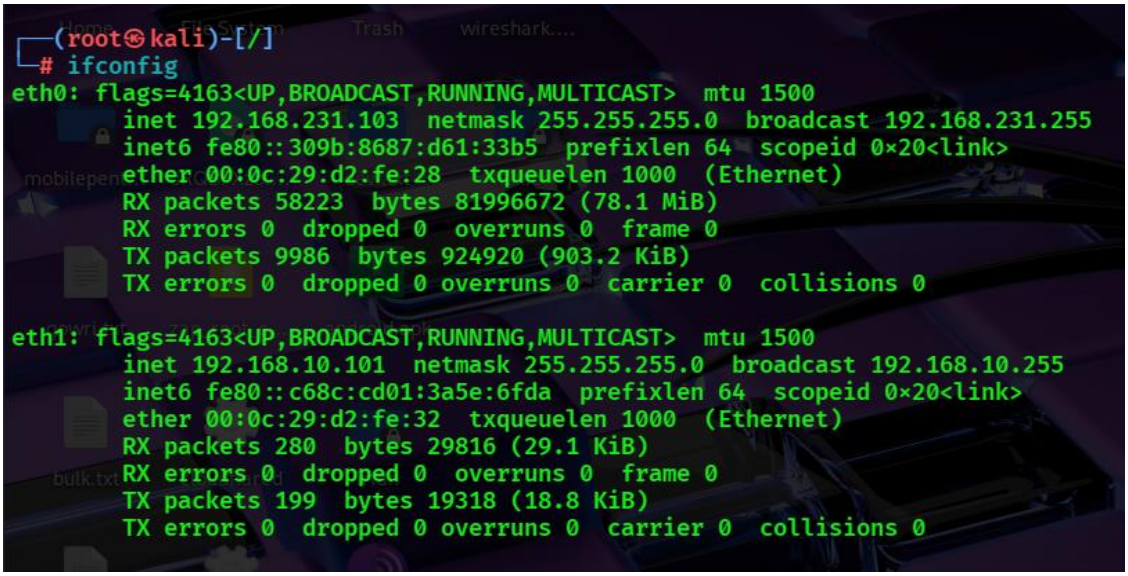
The practical work was completed in the following sequence:

1. Prepared the lab environment. The attacker Kali VM and target Kali VM were connected through the lab network. The target's IP address was recorded as 192.168.231.103, and the attacker's IP address was recorded as 192.168.231.105.
2. Verified SSH connectivity. The SSH service on the target was checked, and connectivity from the attacker VM was confirmed.
3. Verified log collection. SSH authentication events from the target were checked in Splunk before the simulation. This confirmed that the relevant log source was searchable.
4. Performed the controlled simulation. Hydra was run from the attacker VM against the designated target VM in the lab. The simulation was limited to the project environment.

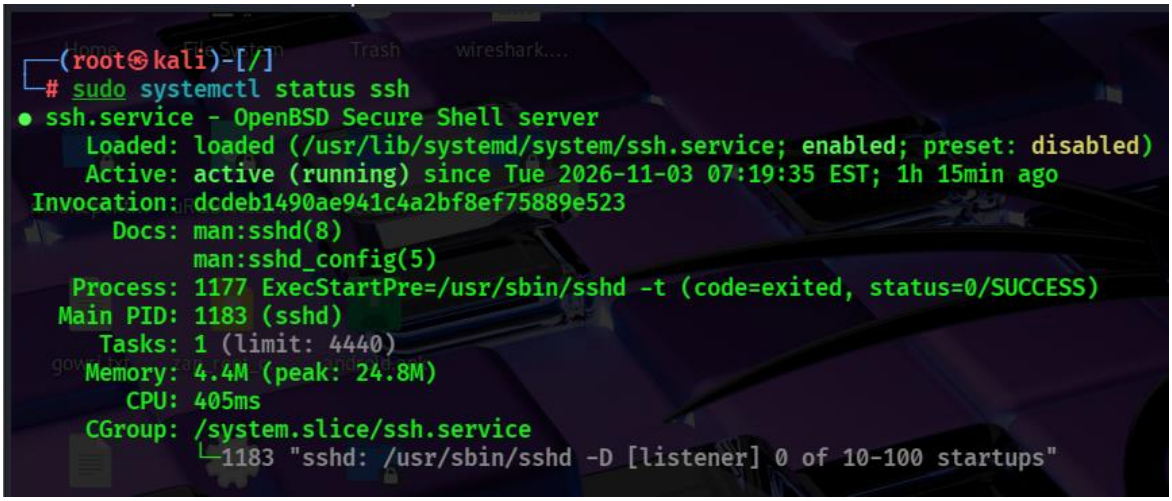
5. Observed the target logs. The target's SSH authentication log was reviewed to check that the login attempts were recorded.
6. Searched the events in Splunk. The collected events were searched and reviewed to identify failed login activity and examine the available fields.
7. Documented the evidence. Screenshots were captured from the virtual machines, the target's authentication log, and Splunk. These screenshots are included in the relevant sections of this report.



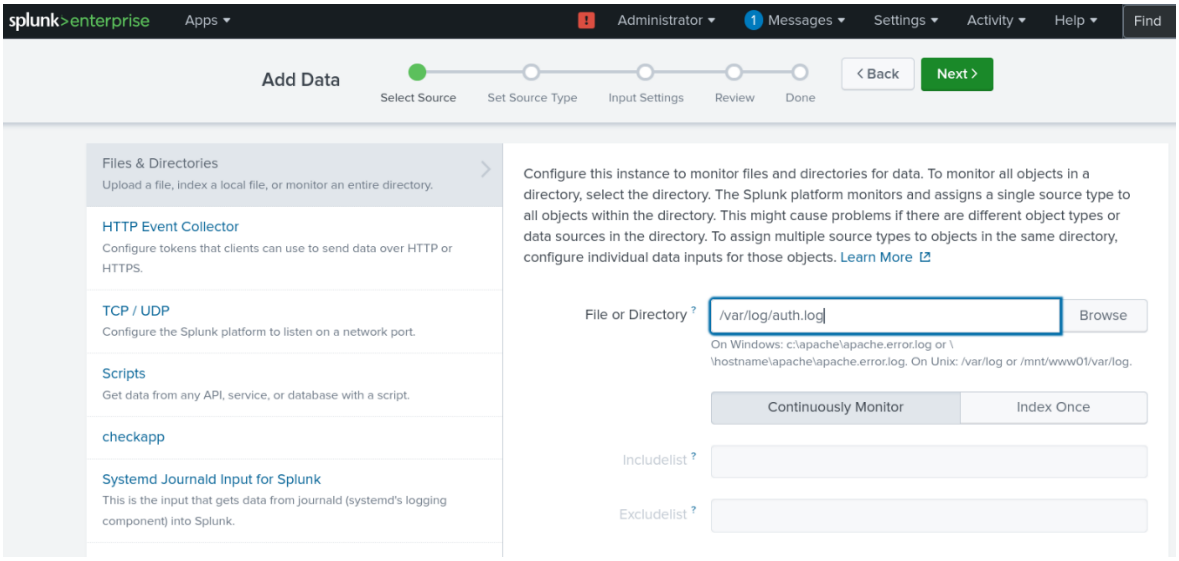
IP ADDRESS OF ATTACKER MACHINE



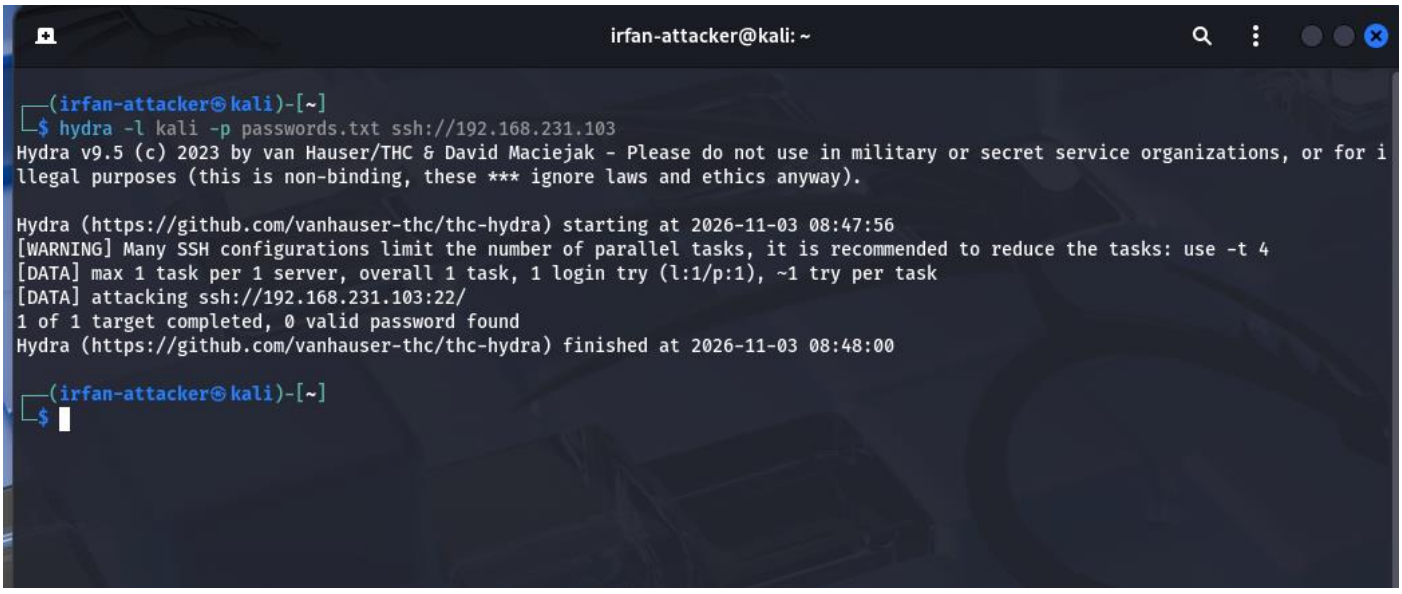
IP ADDRESS OF TARGET MACHINE



SSH SERVICE RUNNING ON TARGET MACHINE



TARGET MACHINE AUTH.LOG INGESTED INTO SPLUNK



HYDRA SIMULATION OUTPUT

SPL Queries and Explanations

The following SPL queries were used during the investigation to search and analyse SSH authentication events collected from the target Kali Linux machine. These searches helped identify failed login attempts, determine their source IP addresses, examine attempted usernames, and analyse the frequency and timing of authentication failures.

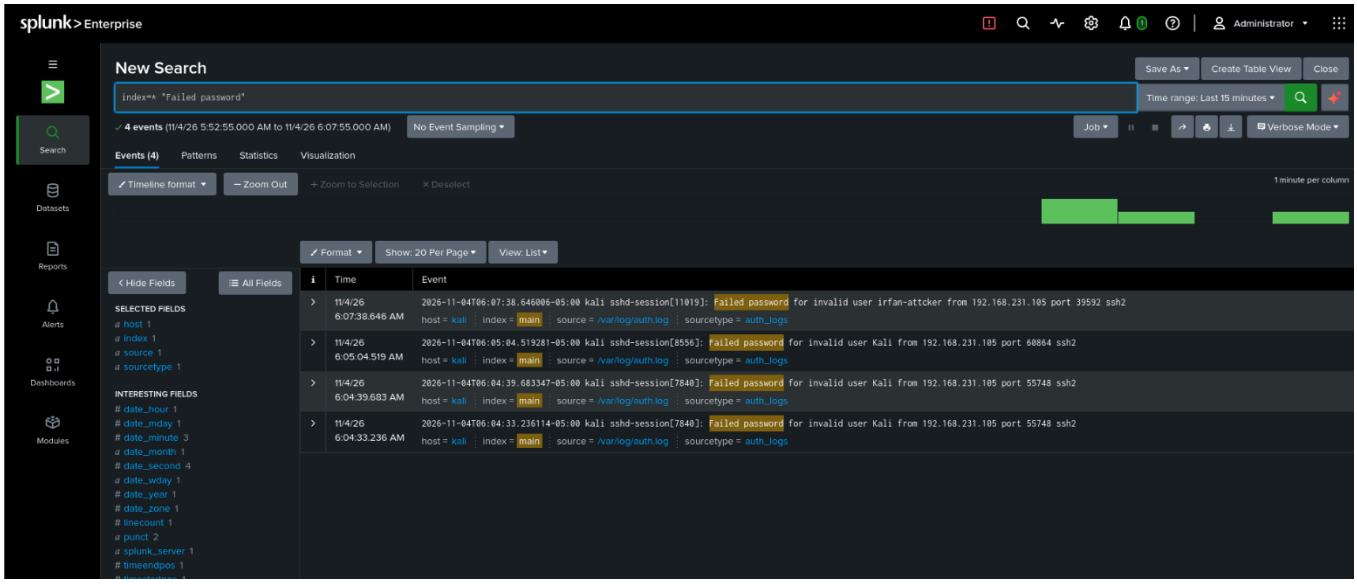
A. Search for failed SSH authentication events

Purpose: Find events indicating failed SSH password authentication.

index=* "Failed password"

Explanation: The index=* searches across all indexes accessible to the user, while "Failed password" filters for events containing that phrase.

Result: Splunk returned **4 events** matching events for the selected time range **Last 15 minutes**.



B. Count matching events

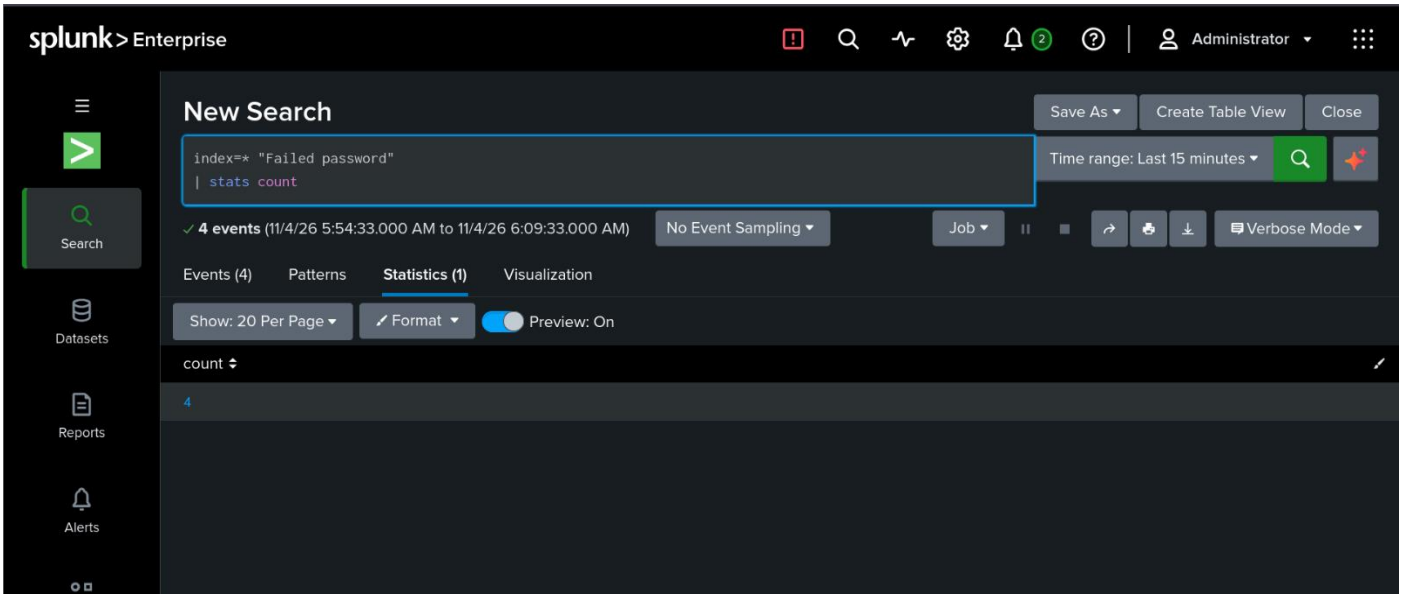
Purpose: Count the failed SSH authentication events found by the search.

index=* "Failed password"

| stats count

Explanation: stats count returns the number of events matching the search.

Result: The search returned 4 matching events.



C. Group failed attempts by source IP

Purpose: Identify which source IP addresses generated the matching events.

index=* "Failed password"

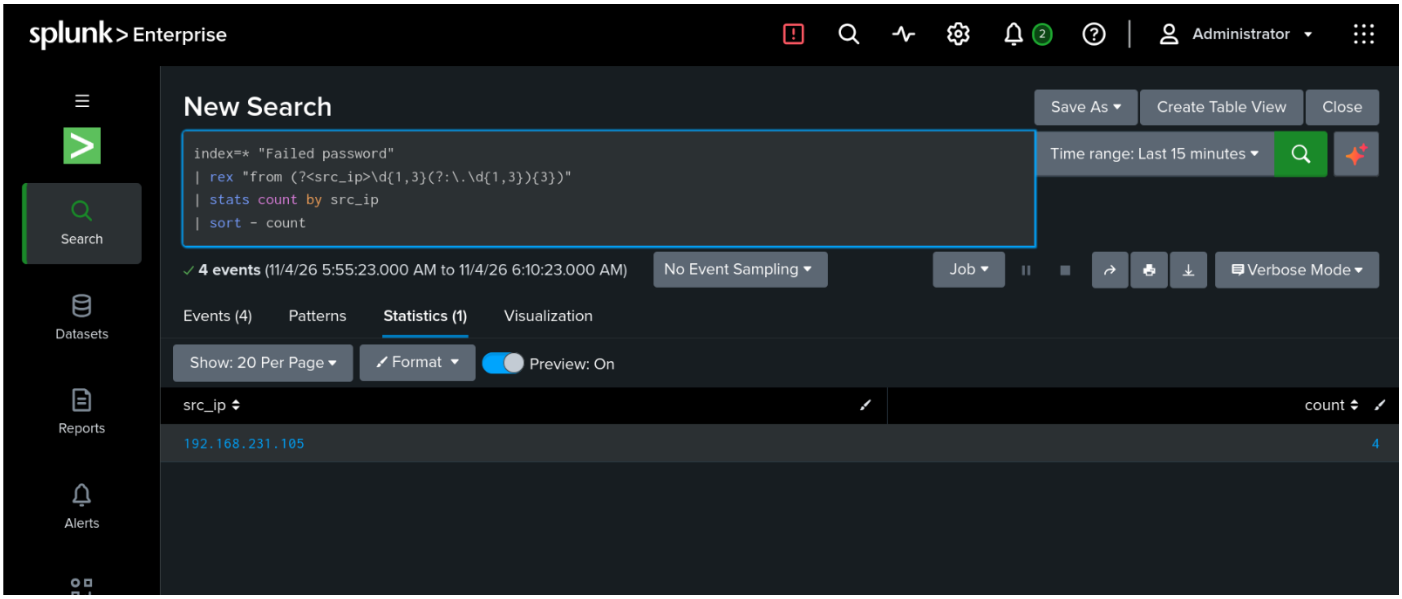
| rex "from (?<src_ip>\d{1,3}(?:\.\d{1,3}){3})"

| stats count by src_ip

| sort - count

Explanation: rex extracts the IPv4 address following the word from. stats count by src_ip counts events for each extracted address, and sort - count orders the results from highest to lowest.

Result: The search showed **source IP: 192.168.231.105, Count : 4.**



D. Group events by username

index=* "Failed password"

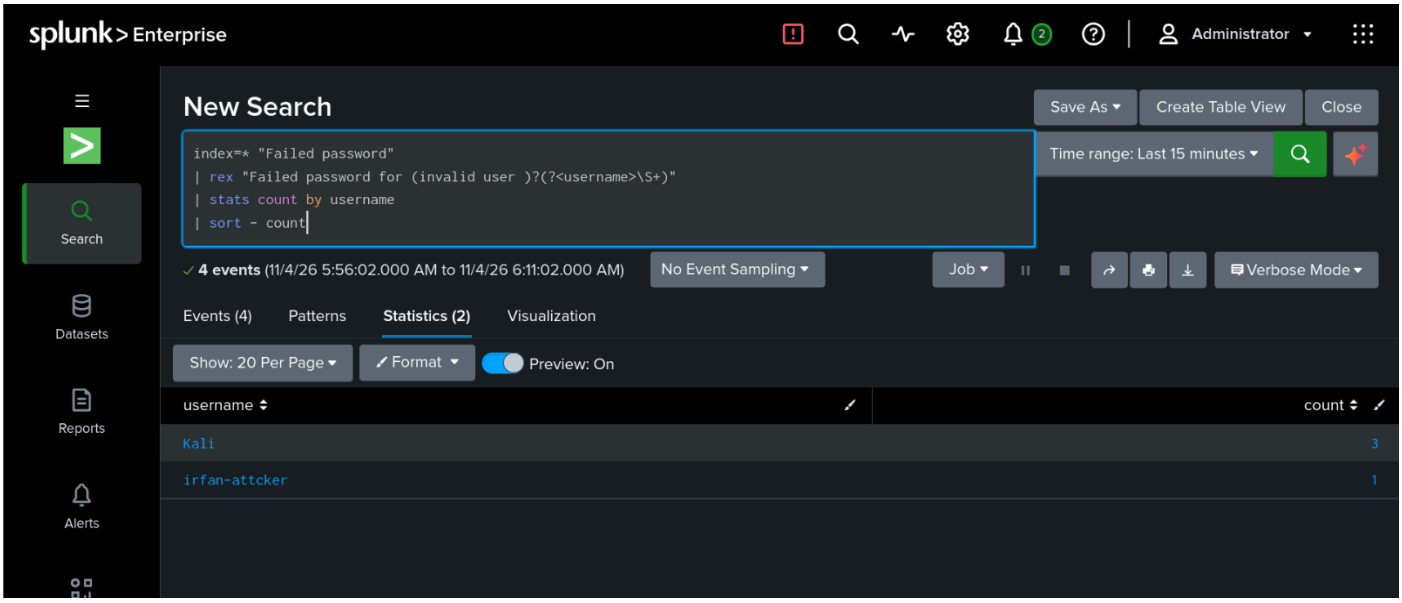
| rex "Failed password for (invalid user)?(?<username>\S+)"

| stats count by username

| sort - count

Purpose: Review the usernames associated with matching failed-login events.

Result: Kali : 3 & irfan-attcker : 1.



E. Examine the event timeline

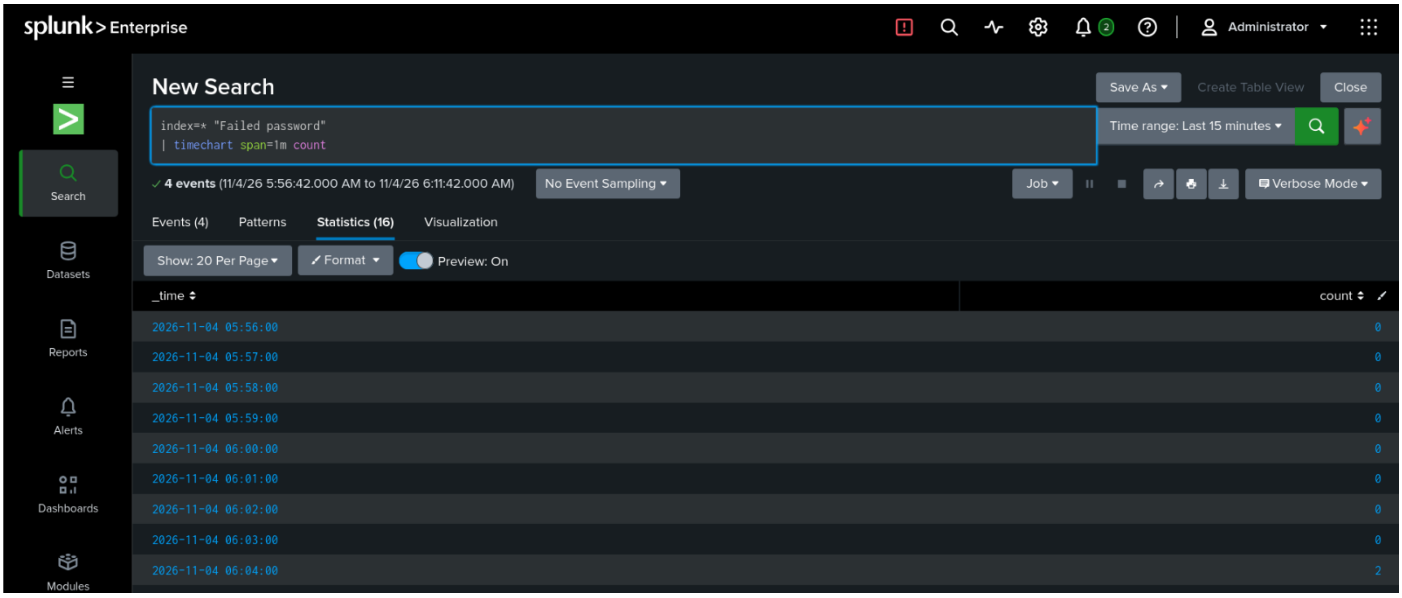
index=* "Failed password"

| timechart span=1m count

Purpose: Review how the matching events were distributed over time.

Explanation: Timechart counts events in one-minute intervals. Confirm that Splunk assigned accurate timestamps before interpreting the chart. The timeline groups matching events into one-minute intervals.

Result: The timeline showed **2026-11-04 05:56:00**.



Results and Analysis

The investigation examined SSH authentication events generated during a controlled lab simulation. The events were collected from the target Kali Linux machine and analysed using Splunk Enterprise.

For the selected search period, Splunk returned four matching failed-authentication events. The source IP analysis identified 192.168.231.105 as the source of all four events. The username analysis identified two usernames associated with the failed login attempts: kali and irfan-attcker.

The timeline analysis was used to examine when the matching events occurred. These findings demonstrate how Splunk can help SOC analysts search, group and investigate SSH authentication failures.

The observed activity was consistent with repeated SSH login attempts during the controlled simulation. However, the results are limited to the collected log data and do not independently establish successful unauthorized access or system compromise.

Detection Logic

The investigation treated repeated failed SSH authentication events from the same source within a short period as activity that may warrant review.

The detection approach was:

1. Search for failed SSH authentication events.
2. Group matching events by source IP, where that field was available.
3. Examine the number and timing of events from each source.

4. Review any successful authentication events near the same time, if present in the collected logs.

The detection is an investigative signal rather than proof of an attack. A legitimate user may also generate failed logins, and a reliable alert threshold should be selected and tested against the environment's normal activity.

Lab rule used: "No fixed threshold was configured; the events were reviewed manually."

Recommendations

Based on the scope of this lab, the following defensive practices are recommended:

- Monitor repeated SSH authentication failures and review unusual concentrations of events.
- Review successful logins that occur after a sequence of failures.
- Restrict SSH access to approved systems and users where practical.
- Prefer SSH key-based authentication when appropriate for the environment.
- Keep SSH and operating-system software updated.
- Tune detection thresholds using normal activity from the environment to reduce unnecessary alerts.
- Retain and review authentication logs so that investigations can be supported by evidence.

Limitations

- The project was performed in a controlled virtual lab and does not represent all conditions in a production environment.
- The analysis depends on the authentication events collected and the fields available in Splunk.
- If source IP addresses, usernames, or timestamps were not extracted correctly, analysis using those fields may be incomplete.
- Repeated failed logins can have legitimate causes; they do not independently prove malicious intent or successful compromise.
- The results apply to the test period and dataset examined in this project.

Conclusion

This project provided practical experience in collecting and analysing SSH authentication logs using Splunk. A controlled Hydra simulation was performed from an attacker Kali VM against a designated target Kali VM. The target's authentication events were then reviewed in Splunk using searches and summaries.

The investigation demonstrated how log searches and event grouping can help identify repeated failed SSH login activity for further review. The conclusions are limited to the evidence collected during the lab. Future work could include refining the detection threshold, validating the searches against additional log samples, and testing alerting and reporting workflows.

References

- **Splunk documentation :** https://docs.splunk.com/Documentation/Splunk/latest/SearchReference/WhatsInThisManual?
- **Splunk search Tutorial :** https://docs.splunk.com/Documentation/Splunk/latest/SearchTutorial/WelcometotheSearchTutorial?
- **Kali Linux documentation :** https://www.kali.org/docs/?
- **OpenSSH Manual Pages :** https://www.openssh.com/manual.html?